

Parallel Voxelization of Triangle Meshes

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Abstract—This paper presents a parallel implementation of conservative voxelization for triangle meshes. The main implementation uses MPI to distribute triangles across processes, build private bit-packed voxel grids, and merge the final result with a bitwise OR reduction. To justify the selected paradigm, two shared-memory OpenMP variants are also implemented: one with a private grid per thread and one with a shared grid updated through atomic OR operations. The evaluation compares execution time, speedup, efficiency, and experimental overhead for small and moderate grid resolutions.

Index Terms—voxelization, MPI, OpenMP, parallel computing, overhead, PRAM

I. INTRODUCTION

Voxelization is the process of discretizing a 3D object into an axis-aligned grid of volumetric pixels. This representation is used in computer graphics, medical imaging, urban analysis, simulation, and interactive applications. This project studies triangle-mesh voxelization as a parallel computing problem: given a set of triangles and a fixed grid resolution, mark the voxels occupied by the mesh.

The first version of the project modeled the algorithm in PRAM and identified task parallelism over triangles. This final version keeps that idea but implements and measures three concrete CPU variants: MPI, OpenMP with private grids, and OpenMP with atomic updates. CUDA and tile-based voxelization are discussed as future work because they require additional GPU memory management or spatial indexing to be correct and measurable.

II. ALGORITHM

A. Input and Representation

The input is a triangle mesh $T = \{t_0, t_1, \dots, t_{m-1}\}$ and a grid resolution r . The voxel grid has $V = r^3$ cells. To reduce memory, the grid is stored as a bitset implemented with 64-bit words, so the number of stored words is $w = \lceil V/64 \rceil$.

For each triangle, the current implementation computes its axis-aligned bounding box (AABB) in grid coordinates and marks every voxel inside that box. This is conservative: it can mark false positives, but it is simple, deterministic, and exposes parallel work clearly.

B. PRAM Model

The high-level algorithm can be modeled as CRCW PRAM because different workers may write to the same voxel. The write conflict is benign when the operation is a logical OR, as marking the same voxel multiple times still leaves the final bit set to one.

Algorithm 1 Voxelization (PRAM)

```
1:  $B \leftarrow$  empty bitset
2:  $T \leftarrow \text{triangulate}(\text{model.load}(\text{path})).\text{faces}$ 
3: for all  $t$  in  $T$  parallel do
4:   for all candidate voxel  $v$  inside AABB of  $t$  do
5:      $i \leftarrow \text{index}(v)$ 
6:      $B[i] \leftarrow B[i] \vee 1$ 
7:   end for
8: end for
9: return  $B$ 
```

c_i is defined as the number of candidate voxels visited for triangle t_i . Then, $C = \sum_{i=0}^{m-1} c_i$. The sequential work of the voxelization stage is:

$$T_s = \Theta(m + C)$$

With unbounded PRAM processors over triangles, the span is dominated by the largest triangle AABB:

$$T_\infty = O(\max_i c_i)$$

Using Brent's theorem, an ideal parallel bound is then:

$$T_p = O\left(\frac{m + C}{p} + \max_i c_i\right)$$

This bound explains the main limitation of triangle partitioning: if a few triangles cover very large AABBs, load balancing can degrade.

III. PARALLEL IMPLEMENTATIONS

A. MPI

The MPI implementation is the main distributed-memory version. Rank zero loads the mesh and distributes triangles with `MPI_Scatterv`. Then, each rank builds its own local bitset grid. The final grid is obtained with `MPI_Reduce` and `MPI_BOR`.

$$T_p^{\text{MPI}} \approx \frac{C}{p} + T_{\text{scatter}}(m, p) + T_{\text{reduce}}(w, p)$$

This design avoids data races during voxel marking, as each process writes to private memory only. Its main overhead comes from process initialization, triangle distribution, synchronization, and reduction of the bit-packed grid.

B. OpenMP with Private Grids

The `omp_private` variant uses one process and multiple threads. Each thread writes to a private grid, and all grids are merged with an OR reduction:

$$T_p^{\text{OMP-private}} \approx \frac{C}{p} + T_{\text{reduce_mem}}(w, p)$$

This version avoids atomic operation overhead, but it increases memory usage by a factor that is close to the number of threads.

C. OpenMP with Atomic Updates

The `omp_atomic` variant uses a shared bitset grid. Each update uses an atomic OR on the 64-bit word that contains the target voxel bit:

$$T_p^{\text{OMP-atomic}} \approx \frac{C}{p} + AC_{\text{atomic}} + C_{\text{contention}}$$

This version saves memory, but several threads can contend for the same 64-bit word even when they update different voxel bits.

IV. EXPERIMENTAL METHODOLOGY

The measured region is `voxel_seconds`. For each mode, the baseline T_s is the same mode with $p = 1$. The metrics are:

$$S(p) = \frac{T_s}{T_p}, \quad E(p) = \frac{S(p)}{p}, \quad (1)$$

$$T_o(p) = pT_p - T_s, \quad R_o(p) = \frac{T_o(p)}{T_s}. \quad (2)$$

Local experiments used the OBJ model `blocky_creeper_like`, grid resolutions 32, 64, and 128, $p \in \{1, 2, 4\}$, and three repetitions per configuration. Table I reports the median for resolution 128, where the parallel work is large enough to expose speedup in some modes.

TABLE I
LOCAL RESULTS FOR RESOLUTION 128, MEDIAN OF THREE RUNS.

Mode	p	T_p (s)	$S(p)$	$E(p)$	O/T_s
MPI	1	0.002703	1.000	1.000	0.000
MPI	2	0.001755	1.540	0.770	0.299
MPI	4	0.002288	1.181	0.295	2.386
OMP atomic	1	0.003745	1.000	1.000	0.000
OMP atomic	2	0.002837	1.320	0.660	0.515
OMP atomic	4	0.002061	1.817	0.454	1.201
OMP private	1	0.003016	1.000	1.000	0.000
OMP private	2	0.002748	1.098	0.549	0.822
OMP private	4	0.002948	1.023	0.256	2.910

V. RESULTS

For resolutions 32 and 64, the measured times are very small and overhead dominates. At resolution 128, MPI obtains its best local point at $p = 2$. OpenMP atomic improves up to $p = 4$ in this small mesh, while OpenMP private does not clearly outperform the other variants because the private-grid reduction is significant relative to the computation.

VI. KHIPU VALIDATION

The project was also validated on UTEC's Khipu cluster using Slurm, OpenMPI, and CMake. The code was adjusted so it can build without Assimp by using a small OBJ fallback loader for benchmark models. A conservative debug job used up to four CPUs and small resolutions. This run is treated as portability validation, not as a definitive benchmark, because the measured times are in the microsecond to millisecond range and are therefore sensitive to scheduler and environment noise.

One operational issue was observed: the available OpenMPI environment did not work correctly with direct `srun` execution for this program. The stable approach was to reserve resources with Slurm and use `mpirun` inside the allocation.

VII. DISCUSSION

The results show that increasing p does not always reduce execution time. For strong scaling, the problem size is fixed, so local computation decreases approximately as B_c/p , but reduction, synchronization, launch, and memory costs do not disappear. Therefore, the best p is not necessarily the largest available p .

MPI is defensible as the main implementation because it represents distributed memory directly, avoids concurrent writes through private grids, and expresses the final merge as a natural bitwise OR reduction. OpenMP is simpler on one machine and can have lower overhead, but its variants expose different costs: private grids spend memory and reduction bandwidth, while atomic updates can suffer contention.

The main weakness of triangle-based partitioning is load balance. If a mesh has only a few large triangles, some workers may receive much more work than others. A tile-based algorithm could assign disjoint voxel regions to workers and eliminate the final OR reduction, but a naive tile implementation would compare too many triangles against too many

voxels. Its work is $\Theta(mr^3/p)$ per parallel step without spatial filtering, while the current triangle-based implementation visits only $C = \sum_i c_i$ candidate voxels. For example, a mesh with 69,451 triangles at $r = 512$ would imply about 9.3×10^{12} triangle-voxel tests in the naive all-triangles-per-tile approach. Practical tile-based voxelizers need spatial filtering structures such as binning, BVH, octrees, or sparse grids.

A. Why CUDA Was Not the Main Delivery

CUDA is technically attractive, especially when a local GPU is available, but the limiting issue is not only whether a kernel can be launched. A correct GPU version needs a memory layout, atomic or block-local reductions for bit updates, host-device transfer timing, and output validation against the CPU implementation.

The memory cost also grows quickly for dense grids. A bit-packed grid requires $r^3/8$ bytes, while a byte-per-voxel grid requires r^3 bytes. At $r = 4096$, the bit-packed grid already needs 8 GiB, and a byte grid needs 64 GiB. If each GPU block or worker stores four private bit-packed grids before reducing, the same $r = 4096$ case needs 32 GiB, above a 12 GiB local GPU. With 12 GiB of VRAM, the theoretical maximum dense bit-packed grid is about $r = 4688$ for a single grid, about $r = 2953$ with four private grids, and about $r = 2344$ with eight private grids, before counting triangles, auxiliary buffers, CUDA runtime overhead, and temporary reductions.

Therefore, a CUDA implementation is reasonable future work only if paired with a carefully chosen memory strategy and a validation plan. For this delivery, MPI and OpenMP provide a simpler and more defensible comparison of distributed and shared memory overheads.

VIII. CONCLUSION

The final implementation keeps MPI as the main solution and adds OpenMP variants for comparison. The experiments confirm that overhead is central to the analysis: for small grids, parallelism can lose to overhead, while for larger fixed grids some configurations obtain speedup. Future work should replace the AABB marking approximation with a triangle-box test, implement tile-based partitioning with spatial filtering, and evaluate CUDA only when GPU execution can be validated against the CPU outputs.

USE OF AI

OpenAI ChatGPT/Codex was used as an assistant to organize the report, review the complexity derivation, compare MPI/OpenMP/CUDA tradeoffs, and prepare documentation. The code, formulas, and conclusions were checked against the repository implementation, official MPI/OpenMP/Khipu documentation, and the technical references below. AI was not used as a primary source for experimental results.

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